

CRUCIBLE: A Multi-Agent Framework for Automated Annotation Rubric Calibration via Iterative Inter-Judge Disagreement Resolution

Bidit Das

Independent Researcher
Plano, Texas, USA
biditdas18@gmail.com

Abstract

Annotation quality is a limiting factor in supervised learning pipelines that rely on the LLM-as-judge paradigm. This paper’s central finding is that an automated convergence criterion can certify agreement that chance-corrected measurement does not support, and that the certification is an artifact of the agreement metric being optimized. Our weighted agreement statistic has a floor of 0.667 at zero unanimity, so a convergence threshold of $\tau = 0.80$ is satisfied at 8 of 20 unanimous items: the loop halts and declares success while 60% of items still carry judge disagreement. The domain that crossed τ on this metric at 81.7% has a chance-corrected inter-judge agreement of only $AC1 = 0.286$. We reach this finding while building CRUCIBLE, a multi-agent system that automatically calibrates annotation rubrics through iterative, coordinator-mediated disagreement diagnosis. No human is in the loop after initialization, and no human reference labels are used during calibration. Applied to 60 YouTube transcripts across three domains, CRUCIBLE reached 96.7%, 81.7%, and 70.0% weighted inter-judge agreement, corresponding to chance-corrected $AC1$ of 0.929, 0.286, and -0.080 . An independent human validation study covering the full corpus—20 transcripts per domain, scored by two disjoint cohorts of three annotators each—finds that judge and human chance-corrected reliability rank the three domains identically, so the judges find difficult what humans find difficult. We report as a negative result that agreement between human consensus and the system’s labels carries almost no evidential weight in this design: five of six domain-round cells fail to beat a majority-class baseline that labels every item with the domain’s dominant class, and the pooled rate of 75% is below the corresponding baseline. Against that baseline, a between-domain contrast we previously reported as preliminary evidence of a human-alignment dissociation does not survive. Two further interventions—porting the calibrated rubric to a frontier judge, and adding a reasoning step before the label—each move the optimized metric without moving agreement with the human reference, and a reasoning step alone moves it by +11.6 points. These results locate CRUCIBLE’s value not in raising raw agreement, and not in match against human consensus, but in diagnosing whether residual disagreement is rubric-fixable, capability-bounded, or prevalence-dominated—and in the accompanying warning that a stopping rule defined on an uncorrected agreement metric will terminate before agreement has been achieved. For a safety-critical

evaluation pipeline, this is the risk-relevant contribution: an operator reading 81.7% agreement as convergence is reading a number whose floor is 66.7%.

Code, data, rubrics, consolidated human ratings and reproduction script —

<https://github.com/biditdas18/rubric-calibration-agent>

Introduction

The LLM-as-judge paradigm has become a standard component of annotation pipelines where human labeling is expensive, slow, or biased (Zheng et al. 2023), but single-model annotation carries systematic biases rooted in training distribution (Thakur et al. 2024). Multi-model ensembles reduce this risk but introduce a new dependency: the rubric must be precise enough that independent models reach consistent conclusions—a poorly specified rubric produces disagreement because different models resolve underspecified criteria differently, not because content is genuinely ambiguous.

This paper originates from a concrete failure of that kind, encountered while building a companion Signal-to-Noise Ratio (SNR) classifier for YouTube educational transcripts (Das 2026). An earlier pipeline—a differently-scoped pilot using a domain-agnostic rubric and a frontier chat-model ensemble—showed substantial and inconsistent disagreement across judges. Manually labeling a gold dataset per domain was rejected as time-intensive and subject to annotator-specific bias reducible only through multiple annotators and adjudication. We therefore designed CRUCIBLE: a coordinator-mediated, iterative calibration loop requiring no human reference labels beyond an initial rubric draft. Three heterogeneous local models serve as judges; a frontier coordinator diagnoses their disagreement patterns and proposes targeted rubric revisions; the loop repeats until convergence or a fixed iteration budget is exhausted.

Building that loop surfaced this paper’s central finding, which concerns the criterion that decides when it has succeeded: a threshold set on an uncorrected agreement statistic is satisfied at a level of actual agreement fixed by that statistic’s floor, so the loop can report convergence while most items remain contested. A human validation study spanning the full corpus supplies the external check that agreement-only pipelines never run.

The core contributions of this paper are:

1. A convergence threshold defined on an uncorrected agreement metric terminates calibration before agreement has been achieved. Our metric has a floor of 0.667, so $\tau = 0.80$ is met at 40% unanimity; Technology & AI’s “converged” 81.7% is a chance-corrected AC1 of 0.286. Two further interventions—a reasoning-before-label ablation and a rubric-portability test—each raise the optimized metric without raising agreement with the human reference, so the metric’s movement is not by itself evidence of improvement. Both score against the round-1 reference set, so they test robustness to judge model and prompt structure rather than adding independent evidence.
2. An automated rubric calibration loop, driven solely by inter-judge disagreement with no human reference labels used during calibration, made tractable on consumer hardware via a columnar evaluation order.
3. A diagnostic framework distinguishing rubric-fixable disagreement from capability-bounded disagreement—a distinction with direct bearing on whether a deployed automated-judge pipeline’s residual error is correctable or a standing limit that risk assessment must budget for.

Related Work

LLM-as-judge evaluation, formalized by Zheng et al. (2023) (MT-Bench), extends to structured rubrics (Liu et al. 2023), multi-dimensional scoring (Kim et al. 2024), and customizable dimensions (Fu et al. 2023); single-model judges show systematic biases (Zheng et al. 2023; Thakur et al. 2024), and multi-model ensembles (Chan et al. 2023) mitigate this but still depend on rubric quality. Rubric-generation and rubric-revision systems—LLM-Rubric (Hashemi et al. 2024), AdaRubric (Ding 2026), Autorubric (Rao and Callison-Burch 2026) and related work (Shen et al. 2026; Siro, Aliannejadi, and Aliannejadi 2026)—either treat the rubric as fixed or validate revision against human-labeled reference data (Shi et al. 2025); rubric-scored instruction-following benchmarks (Su et al. 2026) likewise take the rubric as given. Our narrower claim: we revise domain-specific rubrics using cross-model disagreement as the sole feedback signal, with no human reference labels used during calibration, and ask when that signal certifies agreement it has not achieved—a question reference-validated systems cannot pose. Multi-agent debate (Du et al. 2023) addresses disagreement by exchange without producing generalized rubric revision; CRUCIBLE applies the coordinator-mediator pattern asymmetrically, complementary to weak supervision (Ratner et al. 2017; Zhang et al. 2022) and distillation (Hinton, Vinyals, and Dean 2015).

On the trustworthy-AI side, this work sits with verification and validation approaches that treat an evaluation pipeline’s own reliability as an object of study; our contribution is a concrete diagnostic for deciding when a judge panel’s convergence is evidence of correctness rather than of a permissive threshold.

The CRUCIBLE System

Problem Statement

Given a corpus $D = \{d_1, \dots, d_N\}$ to be annotated with a binary label $y \in \{H, L\}$, and an initial rubric $R^{(0)}$ specifying criteria for each label, we seek a revised rubric $R^{(k)}$ such that the weighted inter-judge agreement $A(R^{(k)}, D) \geq \tau$ for a target threshold τ , with the revision process requiring no human-labeled reference data.

The key insight is that inter-judge disagreement encodes rubric underspecification: when two competent models applying the same rubric reach opposite conclusions on the same document, the rubric has not resolved the case. A coordinator with visibility into both decisions and the criterion-level justifications can identify which criterion was misapplied and propose a targeted revision.

Architecture

Three local judge models (Llama 3.1 8B, Mistral 7B, Qwen 2.5 7B, each at temperature 0.0) independently label every transcript against the current domain rubric $R_d^{(k)}$, in isolation from one another. Their outputs feed a weighted inter-judge agreement score A (Eq. 1); if a domain is below threshold and has not stagnated, a coordinator (Claude Sonnet 4.6, Anthropic 2026) diagnoses the disagreement from per-model criterion quotes, flags any irreducible cases, and proposes a targeted rubric revision that feeds back into the next iteration. The loop stops when all domains converge or after 10 iterations, emitting the calibrated rubrics, labels, and a summary log. Figure 1 gives the full architecture.

Judge models. Three heterogeneous local models via Ollama at temperature 0.0, chosen to represent different training lineages, maximizing the diagnostic value of their disagreements. These serve as the calibration instrument only; production labeling in the downstream application uses a separate, stronger frontier ensemble (Das 2026).

Coordinator model. Claude Sonnet 4.6 (Anthropic 2026), via API. Receives the current rubric, per-model label assignments and criterion-level justification quotes for each disagreement case, and disagreement counts per criterion; produces a structured revision proposal specifying which criterion to modify.

Agreement metric.

$$A = \frac{1}{N} \sum_{i=1}^N w_i, \quad w_i = \frac{1}{K} \max_{c \in \mathcal{C}} n_{ic} \quad (1)$$

where K is the panel size, $\mathcal{C}$ the label set with $L = |\mathcal{C}|$, and n_{ic} the number of judges assigning label c to item i ; w_i is the panel’s majority fraction on that item, and A is bounded below by $f_{K,L} = \lceil K/L \rceil / K$. This work uses $K = 3$ on a binary label, where every item is either unanimous ($w = 1.0$) or a two-one majority ($w = 0.667$) and $f = 0.667$. Under independent judges the expected value of A is approximately 0.75 at $K = 3$, the chance level for this metric; label skew can raise it further. All-agree rate is tracked separately as a stricter secondary metric.

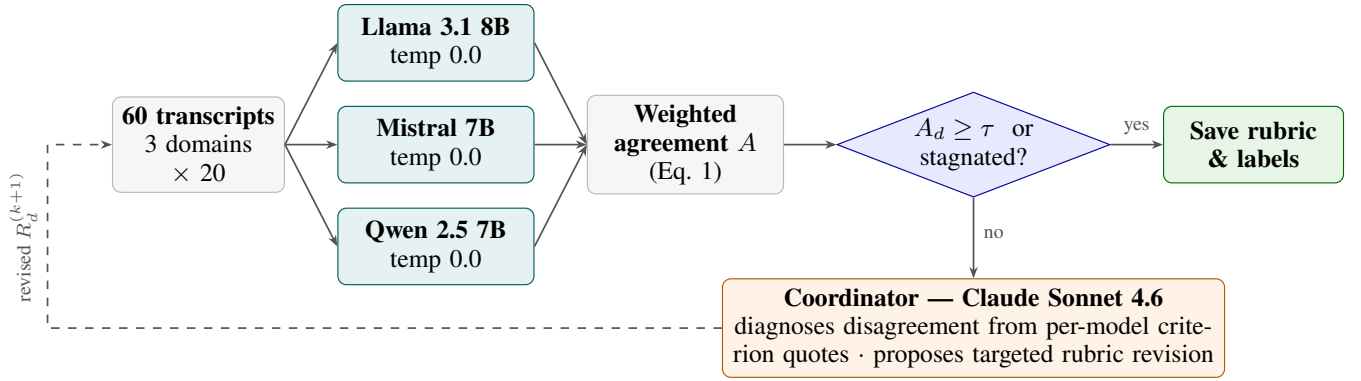


Figure 1: CRUCIBLE system architecture. Three judge models evaluate independently at temperature 0.0. The dashed arrow is the only feedback path: a revised rubric flows back into the next iteration. No human input occurs after initialization.

Calibration Loop

Domain-specific rubrics $R_d^{(0)}$ are written at moderate abstraction. At each iteration: all three judges evaluate all documents; weighted agreement $A_d^{(k)}$ is computed per domain; for each non-converged, non-stagnated domain below τ , disagreement cases are extracted, per-model criterion quotes assembled, and the coordinator proposes a revision applied to the next iteration. A domain is marked stagnated when $\Delta A_d \leq 0$ for three consecutive iterations and held at its current rubric; the loop terminates only when all domains have converged or 10 iterations are reached. A regression check re-evaluates each already-converged domain every iteration.

Judge Quality Control

Prior to the primary run, each candidate judge is screened for discriminative validity. During development, Gemma 2 9B produced LOW labels for all 60 transcripts across all domains (0% HIGH rate)—a degenerate judge providing zero inter-judge signal. It was replaced with Qwen 2.5 7B before the calibration run reported here. This pre-screening gate is itself a risk-relevant safeguard: a degenerate judge inflates agreement without shared understanding, and would silently pass an agreement-only quality check. Separately, once a domain converges, CRUCIBLE re-evaluates it on each subsequent iteration under the same rubric and judges as an output-stability check; no regression (agreement drop > 10 points) appeared in the reported run.

Columnar Evaluation Order

Batching evaluation by model rather than by document (running each judge across all 60 transcripts before switching models) reduces model-load events from 180 to 3 per iteration and cuts per-iteration runtime from an incomplete > 30 minutes to 17–20 minutes on consumer hardware, making the full 10-iteration run tractable. This is an engineering detail, not a scientific finding.

Experimental Setup

Task and corpus. Binary classification of YouTube educational video transcripts under the SNR taxonomy: HIGH

signal (substantive instructional content) vs. LOW signal (generic motivational advice, fear-based framing, promotional material). The corpus comprises 60 transcripts across three domains (20 per domain): General Education, Technology & AI, and Career & Self-Improvement.

Sampling. Transcripts were collected via manual search (~ 20) and the YouTube Data API v3 (~ 130 , spanning this corpus and a 90-transcript held-out supplement), prioritizing a fixed per-domain channel allow-list before falling back to general search—a deliberate diversity-for-consistency trade-off (Limitations).

Target threshold. $\tau = 0.80$: a modest margin above the metric’s ≈ 0.75 chance level under independent judges, not a claim that 0.80 represents strong agreement. We treat τ as a practical stopping rule; chance-corrected metrics (Fleiss’ κ , Fleiss 1971; Gwet’s AC1, Gwet 2008) are what we rely on to judge genuine convergence.

Runtime. 17–20 minutes per iteration, 170.7 minutes total; the run terminated at the maximum-iteration cap of 10 on a consumer MacBook (Apple Silicon, no discrete GPU).

Results

Per-Domain Convergence

General Education converged immediately (iteration 1, no revision). Technology & AI required one revision: the coordinator diagnosed that Qwen 2.5 7B was reading the technical-specificity criterion too narrowly; the revised criterion raised agreement from 76.7% to 81.7%, crossing threshold in one step. Career & Self-Improvement required two revision attempts: the first (v1) reached 70.0%, the second (v2) regressed to 68.4% and was discarded; agreement never approached the 80% target (Table 2). The coordinator’s diagnosis across both iterations consistently identified the same failure mode: one judge applied the “generic motivational language” criterion to any transcript containing motivational phrasing, regardless of co-occurring procedural content.

We report Fleiss’ κ (Fleiss 1971) and Gwet’s AC1 (Gwet 2008) alongside raw weighted agreement because raw agreement is inflated by label prevalence (Table 1). Technology &

Domain	Best A	κ	AC1
General Education	96.7%	-0.034	+0.929
Technology & AI	81.7%	+0.246	+0.286
Career & S-I	70.0%	-0.350	-0.080

Table 1: Calibration results per domain. κ is Fleiss’ κ (Fleiss 1971) and AC1 is Gwet’s AC1 (Gwet 2008), both computed over the three judges’ labels.

Iter.	Career & SI	Tech & AI	Gen. Edu.
1	68.4	76.7	96.7 ✓
2	70.0	81.7 ✓	96.7 ✓
3	68.4 (regressed)	81.7 ✓	96.7 ✓
4–10	68.4 (no further gain)	81.7 ✓	96.7 ✓

Table 2: Weighted agreement (%) per domain across iterations. Career’s second revision (v2) regressed at iteration 3 and was not retained; the final output is v1 (70.0%, iteration 2). A check mark marks a domain at or above $\tau = 0.80$.

AI shows fair chance-corrected agreement—the only domain where a rubric revision carried agreement to the convergence target. Career & Self-Improvement shows below-chance agreement: the judges are systematically anti-correlated relative to that metric’s chance expectation, which no rubric revision could change. General Education shows $\kappa \approx 0$ despite 96.7% raw agreement—the known prevalence paradox of κ under skewed marginals (Byrt, Bishop, and Carlin 1993), with Gwet’s AC1 less affected (Gwet 2008). The practical point: CRUCIBLE’s contribution is not raising raw agreement per se, but diagnosing which regime a domain occupies: rubric-fixable, capability-bounded, or prevalence-dominated. Most of the final agreement is accordingly attributable to the initial rubrics rather than to revision: General Education converged with no revision at all, Technology & AI gained 5.0 points, and Career & Self-Improvement gained 1.6 and never approached τ . This bounds any claim about the calibration loop’s effectiveness as an agreement-raising mechanism, and is consistent with locating its value in diagnosis instead.

The stopping rule is satisfied before agreement is. Equation 1 assigns weight 0.667 to every non-unanimous item, so A has a floor of 0.667 when no item is unanimous. Solving $A \geq \tau$ for the unanimous count u over N items gives $u \geq (\tau - 0.667) N / 0.333$, which at $\tau = 0.80$ and $N = 20$ is $u \geq 8$. The convergence criterion is therefore met when 40% of items are unanimous and 60% still carry judge disagreement. Technology & AI crossed τ with $u = 9$ of 20; General Education, which converged without revision, had $u = 18$. The two outcomes the loop reports identically as “converged” differ by a factor of two in unanimity and by 0.643 in chance-corrected agreement (AC1 = 0.286 vs. 0.929; Table 1). This is a property of the threshold, not of the domains: any stopping rule on an agreement statistic with a non-zero floor halts at a chance-corrected level fixed by that floor, so τ must be set relative to the floor rather than on the unit interval (the

floor-normalised form is given below). We retain A in the trajectory tables because the loop optimized it, and report AC1 beside every value.

Pairwise structure of the panel. Panel-level A also conceals which judges carry it. Llama 3.1 8B and Mistral 7B are the highest-agreeing pair in every domain (100.0%, 75.0%, 90.0%) and Qwen 2.5 7B is the systematic dissenter. Technology & AI crossed τ at the panel level while its Llama–Qwen pair agreed on only 45.0% of items, below that pair’s own chance expectation (AC1 = -0.08): the crossing is a two-of-three majority carried by the more HIGH-leaning pair, not a uniform consensus—a second line of evidence, internal to the judges, that the threshold is permissive. Career’s non-convergence is near-total anti-correlation rather than noise (Llama and Qwen agree on 2 of 20 items), which no rubric revision reachable by this loop could repair, and the prevalence effect recurs pairwise: General Education’s Llama–Qwen pair scores 90.0% raw agreement at Cohen’s $\kappa = 0.00$.

Human Validation

A known limitation of LLM-as-judge pipelines is that inter-judge agreement among automated annotators may reflect shared model bias rather than agreement with independent human judgment. We address this with an independent human validation study covering the entire corpus: 20 transcripts per domain, run in two rounds of 10. Each round recruited three annotators per domain (18 annotators in total, Prolific approval rating above 95%, recruited via Prolific, Palan and Schitter 2018), and the two rounds drew disjoint cohorts—no annotator appears in more than one round or more than one domain. Round 1 scored the even-indexed transcripts of each domain block and round 2 the odd-indexed ones, a systematic rather than random split of the 60-transcript corpus. Annotators saw only the raw transcript text—not the calibrated rubric, domain label, or CRUCIBLE’s output—and judged, with mandatory written justification, whether each transcript was substantive or generic.

Two results follow, and they point in different directions.

The first is a correspondence. Chance-corrected reliability among the human annotators ranks the domains in exactly the order the judge panel does: General Education highest (human AC1 = 0.70 pooled, judge AC1 = 0.929), Technology & AI second (0.33 vs. 0.286), Career & Self-Improvement lowest (0.24 vs. -0.080). With three domains this is a rank statement rather than a tested relationship, but the magnitudes agree closely in the two harder domains, and the direction is unambiguous: the judges find difficult what the humans find difficult. Figure 2 (left) shows both rater panels with their intervals. This correspondence is evidence against reading the judge panel’s failures as arbitrary, and it is the reason we relocate the paper’s central claim from human misalignment to the stopping rule.

The second is a negative result about the match statistic itself. Because each domain’s system labels are heavily skewed toward one class, a rater who assigns that class to every item scores 100%, 60%, and 90% in the three domains respectively. Measured against that majority-class baseline,

Domain	Rd	AC1	vs. CRUC.	Base
Gen. Edu.	1	0.63	90%	100%
	2	0.76	100%	100%
Tech. & AI	1	0.63	50%	60%
	2	-0.03	80%	60%
Career & S-I	1	0.33	50%	90%
	2	0.23	80%	90%
Overall		—	75%	83%

Table 3: Human validation at 20 transcripts per domain, scored by two disjoint cohorts of three annotators (round 1 and round 2). AC1 is Gwet’s AC1 among the three annotators within a domain-round. Pooled over all 20 items, with 95% percentile bootstrap CIs (10,000 resamples over items), AC1 is 0.70 [0.43, 0.89] for General Education, 0.33 [-0.02, 0.64] for Technology & AI, and 0.24 [-0.06, 0.56] for Career & Self-Improvement. “vs. CRUC.” is the share of transcripts where human majority-vote consensus matches CRUCIBLE’s calibrated label. “Base” is the majority-class baseline: the score a rater achieves by labelling every item with the domain’s dominant system-label class. Bold marks the only cell that exceeds its own baseline.

five of the six domain-round cells in Table 3 fail to beat a constant labelling, and the pooled rate of 75% is below the pooled baseline of 83%. The single exception is Technology & AI round 2 (80% against a 60% baseline). General Education’s 100% in round 2 ties a constant all-HIGH rater exactly. We therefore report human–system match as uninformative in this design and do not rest any claim on it.

What the enlarged sample does and does not establish.

An earlier version of this study, at $n = 10$ per domain, reported a between-domain contrast in human–system match (9/10 vs. 5/10) as preliminary evidence of an alignment dissociation, noting it did not reach significance on its own (Fisher exact $p = 0.141$) and that roughly 19 items per domain would be required for 80% power. Doubling to $n = 20$ did not confirm that contrast; it removed its basis. Technology & AI’s match moved from 50% to 80% between rounds, pooling to 65%, and the majority-class correction above shows that General Education’s high match was never distinguishable from constant labelling. We report this as a failed replication of our own earlier result.

On the reliability measures, where the enlarged sample does buy resolution, we state what clears zero. Among the judges, General Education’s AC1 exceeds Technology & AI’s (+0.64, 95% CI [+0.31, +0.94]) and Career’s (+1.01, [+0.73, +1.21]), and Technology & AI over Career is marginal (+0.37, [-0.01, +0.73]). Among the humans, only General Education over Career clears (+0.46, [+0.04, +0.80]); General Education over Technology & AI does *not* (+0.37, [-0.04, +0.77]) and Technology & AI versus Career is indistinguishable (+0.09, [-0.39, +0.52]). Technology & AI’s human AC1 does change between cohorts (+0.66, [+0.03, +1.11]), so its human reliability is cohort-dependent where General Education’s is not. At 20 items and three raters per cohort these intervals are wide; we

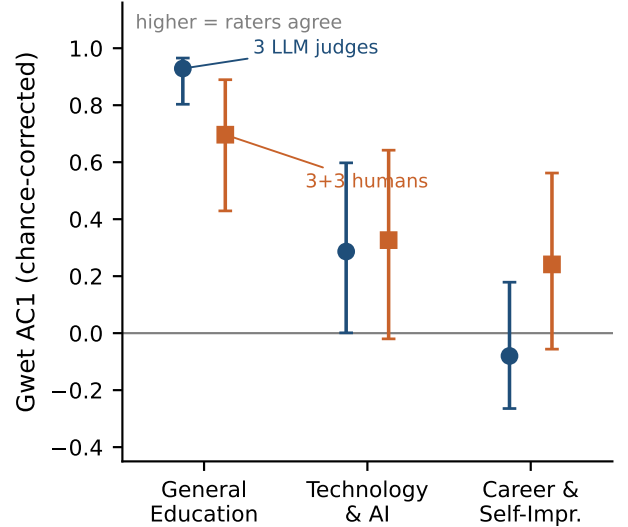


Figure 2: Chance-corrected reliability for the three LLM judges and for the human annotators, by domain. Both rater panels rank the domains identically (Spearman $\rho = 1.00$ over three domains, descriptive only): the judges find difficult what the humans find difficult. Bars are 95% percentile bootstrap intervals over items (10,000 resamples, $n = 20$ per domain per panel). The two domains the calibration loop reports as converged sit at judge AC1 = 0.93 and AC1 = 0.29.

present the estimates as a ranking with stated uncertainty, not as resolved effect sizes.

Scope of the reference set used by later analyses. The rubric-portability test, the reasoning-before-label ablation and the frontier-ensemble comparison below all score against the original round-1 30-item reference and were not re-run at $n = 20$. Their numbers are correct for what they measure, but they inherit the limitation above—human-consensus match is a weak statistic in this design—so they are evidence about consistency across system configurations, not about human alignment.

Automated convergence correctly identified General Education’s annotator group as reliable and Career & Self-Improvement’s as difficult, and the human data now confirm that ordering directly. What it did not do is signal that Technology & AI’s crossing of τ rested on 9 unanimous items out of 20. That is the paper’s risk-assessment finding in its corrected form: the failure is in the stopping criterion, not in the judges’ relationship to human judgment. An operator needs the chance-corrected value and the floor of the metric being thresholded, not the thresholded value alone. We did not recalibrate rubrics toward the human labels, to avoid reintroducing circularity; the study’s role is diagnostic, not corrective.

Judge Replacement

Replacing the diagnosed bottleneck judge with a different model on Career raised inter-judge agreement (0.700 $\rightarrow$

0.883), but the majority-vote label changed on 0 of the 20 transcripts: the two retained judges already agreed 90% of the time, so no third judge could change the output. This is a unanimity effect (2–1 becoming 3–0), not an actual annotation change—a cautionary result for any risk assessment that treats rising agreement as evidence of improving accuracy without checking whether the underlying labels moved at all. This finding is consistent with, but does not independently establish, the paper’s central claim, given a structural confound in the panel tested here (the two retained judges’ pre-existing 90% agreement).

Rubric Portability Across Judge Models

A natural question is why a single frontier model, applied zero-shot, would not suffice in place of the full calibration loop. We test this directly, and in doing so obtain the first of two probes of whether movement in the optimized metric tracks movement in agreement with the human reference. Both probes share that reference, so they test robustness to the judge model and to prompt structure rather than supplying independent evidence.

We compare three approaches against the same 30-transcript human reference used above: (1) the local three-judge panel applying its calibrated rubric; (2) GPT-5.6 Sol (OpenAI 2026) applying a naive zero-shot prompt with no rubric; and (3) GPT-5.6 Sol applying the same calibrated rubric used by the local panel. We use a non-Anthropic model for this baseline because Claude serves exclusively as the calibration coordinator throughout this work and was not evaluated as a judge, to avoid conflating baseline performance with the model family responsible for rubric design. GPT-5.6 Sol does not support temperature=0 at time of writing; results reflect the model’s default sampling temperature.

Against the round-1 human reference ($n = 30$, 10 per domain), the local calibrated panel and a single zero-shot frontier pass achieve identical aggregate accuracy (both 63.3%; per domain 50/50/90% for the local panel and 60/30/100% for GPT-5.6 Sol on Career, Technology & AI and General Education), so the 10-iteration loop does not categorically outperform one frontier zero-shot pass. Giving the frontier model the *same* calibrated rubric does not add value and drops it to 50.0% overall, degrading Career substantially (60% $\rightarrow$ 20%). Comparing the same frontier model’s outputs against the local panel’s labels directly, rather than against human judgment, shows the rubric *does* make GPT-5.6 Sol’s outputs converge toward the local panel’s specific behaviour (agreement with the local panel rises from 0.500 to 0.633 when the rubric is added). Taken together, the calibrated rubric teaches a frontier judge to mimic the local ensemble’s particular behaviour pattern—including the local ensemble’s specific errors—rather than making it more accurate against the human reference. The comparison is within-intervention—the same items scored with and without the rubric—so unlike the between-domain contrast in the Human Validation section it does not rest on per-domain sample size, but it inherits that section’s finding that human-consensus match is prevalence-dominated here. We therefore read it as evidence about what the rubric transfers (the

panel’s behaviour, including its errors), not as a measurement of alignment. It is not a single-draw artifact: repeating both variants five times per transcript (600 calls) gives 87–93% of transcripts an identical label across all five draws, with majority-of-five accuracy matching the single-draw result exactly.

For deployment this means a rubric calibrated against one judge panel should not be assumed portable without re-validation, and rising agreement after transfer is specifically not evidence that the transfer succeeded.

Reasoning-Before-Label Ablation

Motivated by the question of whether Career & Self-Improvement’s stagnation reflects a single-pass prompting limitation rather than a true capability ceiling, we added an intermediate reasoning step to the local judges’ prompts, in the style of chain-of-thought prompting (Wei et al. 2022) (reasoning generated before the label, not after), and re-scored all three domains against their existing calibrated rubrics using the same three local models at temperature=0.0. We call this a *reasoning-before-label prompt* rather than chain-of-thought throughout, since we have no access to the models’ actual internal computation—only to text generated before the label in response to an instruction to reason first.

The reasoning step reliably raises inter-judge agreement in the two domains with room to improve—Career A 0.684 $\rightarrow$ 0.767 (+8.3 points) and Technology & AI 0.817 $\rightarrow$ 0.933 (+11.6 points, with AC1 nearly tripling)—but it never *helps*, and in Career actively *hurts*, agreement with the human reference: Career falls 50% $\rightarrow$ 40% (Cohen’s κ , Cohen 1960, flips +0.14 $\rightarrow$ -0.25), Technology & AI is flat at 50% while its AC1 drops, and General Education is unchanged at 90% while dipping slightly on inter-judge agreement (0.967 $\rightarrow$ 0.917), consistent with a ceiling effect.

This rules out simple single-pass prompting failure as the sole explanation for Career & Self-Improvement’s stagnation—the judges *can* be made to agree more with reasoning added—but shows that forcing reasoning-before-label makes the panel more internally consistent without making that consistency more correct. It is the second probe of whether the optimized metric tracks anything beyond itself: a +11.6-point movement in A leaves agreement with the human reference flat. Like the portability test it scores against the same 30 human labels and inherits their prevalence limitation, so it does not enlarge the evidence base; what it establishes is that A can be moved substantially by a prompt change alone, which is a further reason not to read a threshold crossing on A as evidence of improved annotation. We revise our diagnosis of Career accordingly. The enlarged human study shows that Career’s human annotators agree with each other barely above chance themselves (AC1 = 0.24 pooled, 95% CI [-0.06, 0.56]), the lowest of the three domains and matching the judges’ own ordering. The stagnation is therefore better read not as judges failing to reach a well-defined human standard, but as a domain in which the substantive/generic distinction is weakly determined for judges and humans alike—the capability-bounded cell of our trichotomy, now with rater-side evidence. Explicit reasoning

raises internal consistency without supplying the missing construct. We used a single, untuned reasoning-before-label prompt design across all domains; whether a more carefully engineered prompt would alter this pattern is untested.

Additional Diagnostic Analyses

Four further diagnostic analyses are summarized here, with full designs in the companion technical report (Das 2026).

Held-out generalization. Re-scoring calibrated rubrics on a 90-transcript unseen set improves the weighted inter-judge agreement metric for Career & Self-Improvement (72.3% $\rightarrow$ 78.9%) and Technology & AI (84.5% $\rightarrow$ 87.8%), though the majority-vote annotation changes on 0 of the 90 transcripts—only the metric moves, not any output label. “Held-out” here means document-held-out, not source-held-out: the 60- and 90-transcript corpora share 6 of 46 and 50 channels respectively. The held-out corpus is also 100% majority-vote HIGH in every domain, so this result shows that unambiguous HIGH labels are stable under the calibrated rubric—not that HIGH/LOW discrimination generalizes.

Downstream classifier case study. As a usability case study, not a validation of agreement with the human reference, an SVM trained on calibrated-rubric labels achieved $F_1 = 0.871$ (recall = 0.964) on HIGH-signal, vs. $F_1 = 0.000$ for a majority-class baseline (Das 2026). That study uses the companion report’s separate corpus—89 real transcripts labeled by a frontier ensemble plus 300 synthetic transcripts added for class balance—so its class distribution is not that of the 60-transcript corpus above.

Frontier ensemble vs. human judgment. The separate frontier ensemble that produced the classifier’s training labels tracks human judgment about as loosely as the local panel (56.7% vs. 63.3%; a full-3-judge-subset check gives a consistent 55.0%), so the classifier’s $F_1 = 0.871$ rests on similarly-reliable labels. Both figures are measured against the round-1 reference and inherit its prevalence limitation, so we report them as showing that a stronger, separately-configured ensemble behaves no differently here—not as a measurement of alignment.

Irreducible cases. The coordinator flagged two Technology & AI transcripts as irreducible (genuine ambiguity, not rubric failure) and excluded them from stagnation counting. For this domain, the convergence conclusion (81.7%, above threshold) is identical whether these cases are included or excluded from the denominator. Re-auditing which two transcripts were flagged proved infeasible: the pipeline persists only calibrated rubrics, labels and a run summary, and flagged indices are held in memory only. As an upper bound, 11 of 20 Technology & AI transcripts still show inter-judge disagreement under a full final-rubric relabel—a superset of the two, not those specific cases.

Panel size. Eq. 1 holds for arbitrary K and L , but a fixed τ does not transfer across panel sizes, because $f_{K,L}$ is non-monotonic in K : for a binary label it is 0.500, 0.667, 0.500 and 0.600 at $K = 2, 3, 4, 5$, since even panels admit ties. A fixed $\tau = 0.80$ therefore sits 0.60, 0.40, 0.60 and

0.50 of the way from floor to ceiling at those sizes—the $K = 3$ panel used here is the most permissive of any small panel, not the strictest—and cross- K comparison requires the floor-normalised $A^* = (A - f_{K,L}) / (1 - f_{K,L})$. The judge-replacement diagnostic tests this directly. Across all subsets of the four judges scored on Career & Self-Improvement, raw A ranks the three-judge Llama–Mistral–Granite panel (88.3%) above the two-judge Mistral–Granite panel (87.5%) while A^* reverses them (65.0% vs. 75.0%), with two further inversions. Raw A is therefore interpretable only at fixed K , and τ must be re-derived per panel size. This is demonstrated in one domain at $K \leq 4$; the generalisation itself is exact by construction.

Robustness of the Human Validation Result

Two checks probe the stability of the round-1 human study. Both were computed on that round’s 30 items and were not re-run after the sample was enlarged; they are retained because they bear on the annotator-rejection decision and on channel clustering, neither of which the enlarged sample addresses.

Annotator-rejection sensitivity. One of nine Prolific submissions (Technology & AI) was rejected on a multi-factor judgment (completion time, an attention-check failure, low-effort justifications) and replaced. Recomputing this domain’s human consensus with the rejected annotator’s original responses reinstated in place of the replacement changes zero of the 10 item-level consensus labels. Local-panel-vs-human agreement for Technology & AI is identical under both the official and reinstated consensus (50.0%, Cohen’s $\kappa = -0.19$, Gwet’s AC1 = 0.20), so this domain’s reported divergence does not depend on the rejection decision.

Channel-clustering sensitivity. Resampling at the channel level rather than the transcript level (10,000 resamples, 27 unique channels of 30 items) shifts the round-1 intervals only modestly—the overall upper bound moves 80.0% to 79.3%—so clustering has a small effect on that sample. Source-channel metadata is not retained per transcript in the released corpus file, so the check could not be extended to all 60 items; clustering at that scale (up to 5 transcripts per channel) remains unquantified.

Reproducibility of the Original Run

We recovered the run timestamp, current model digests, quantization (Q4_K_M for all three judges) and context windows via `ollama show`, and confirmed from the judge-calling code that temperature was explicitly 0.0 with no context-window override, ruling out truncation as an explanation for the observed misreadings. We could not recover shell history for the original model pulls, coordinator API logs, or a digest snapshot to diff against, so the follow-up analyses were run under the closest recoverable configuration and should be read as diagnostic extensions rather than exact controlled replications.

Discussion

The Career & Self-Improvement ceiling shows a class of disagreement that rubric revision did not fix under this judge and

prompt setup. An operator seeing 70.0% agreement would not know whether to keep revising the rubric or replace the judge; the stagnation pattern and the coordinator’s consistent diagnosis suggest the rubric was not the limiting factor, though judge replacement could not fully separate this from a structural property of the panel.

For AI risk assessment specifically, the paper’s contribution is narrower and more actionable than we first claimed. It is not that automated judges systematically diverge from human judgment—the enlarged human study gives no support for that, and the reliability correspondence reported above argues against it. It is that a convergence criterion defined on an uncorrected agreement statistic reports success at a level of actual agreement determined by that statistic’s floor. An operator who reads 81.7% as “converged” is reading a number that cannot fall below 66.7% and that corresponds here to a chance-corrected AC1 of 0.286. The deployment risk is not a confidently-wrong evaluator; it is a stopping rule that halts while most items are still contested, and a reported figure that conceals this by construction. The remedy is arithmetic rather than procedural: threshold a chance-corrected statistic, or set τ relative to the floor of the statistic being used.

More broadly, agreement-based confidence bounds (Eq. 1) are informative only conditional on both the panel’s construct validity and the scale of the statistic being thresholded. CRUCIBLE’s three-way split—rubric-fixable, capability-bounded, prevalence-dominated—offers a reusable vocabulary for characterizing residual disagreement, and this paper supplies an instance of the third category applying to a paper’s own headline measurement: the human–system match rates we previously reported as evidence are prevalence-dominated, and we retract them as such.

Limitations

- **Single experiment.** 60 transcripts, three domains; generalization to other domains or document types is untested.
- **Statistical resolution of the human study.** The human study now covers the full corpus at $n = 20$ per domain, which is the sample size an earlier version of this work identified as required. It did not resolve the comparison it was intended to resolve: the human–system match contrast disappeared under the majority-class correction, and on the reliability measures General Education over Technology & AI still does not clear zero ($+0.37$, $[-0.04, +0.77]$). Three raters per cohort is the binding constraint, not the item count; resolving the human-side domain ordering would require more raters per item rather than more items.
- **Non-standard agreement metric.** The calibration loop optimizes A (Eq. 1), whose 0.667 floor is the subject of this paper’s central finding. We report AC1 alongside it throughout, but the calibration trajectories, the stopping decisions, and therefore the released rubrics are all products of the uncorrected statistic. A re-run thresholding AC1 directly would likely produce different rubrics and different per-domain outcomes; we have not performed it.
- **Two-cohort pooling.** The 20 items per domain were scored by two disjoint cohorts of three. Pooled AC1

therefore mixes within-cohort and between-cohort variation, and we report per-round values alongside every pooled figure for this reason. Technology & AI’s cohort-dependent reliability shows the mixing is not negligible.

- **Model-specific ceiling.** The Career ceiling is diagnosed for one judge model (Qwen 2.5 7B) under one untuned prompt template; the reasoning ablation only partially addresses prompt reformulation, so the ceiling should not be called inherent to the model.
- **Human-validation design.** Unaided judgment only; a rubric-guided condition would additionally test whether humans agree with the rubric’s operational criteria rather than with intuition. Each domain also drew a distinct annotator set, so domain and cohort effects cannot be fully separated.
- **Coordinator dependency and run reproducibility.** A non-bit-reproducible frontier coordinator gives an estimated ± 1 –2 point run-to-run variance (unverified, single run); Career’s iteration-2 gain falls within that band. Model digests, quantization and context window were recovered from available logs and are consistent with, but not proof of, the original configuration; coordinator API logs were not accessible.
- **Baselines and judge replacement.** No baseline beyond a single frontier judge was run against manual rubric revision or coordinator-free self-refinement, and judge replacement was tested once on a domain where the two retained judges already agreed 90% of the time.

Ethics Statement

Human validation used Prolific (Palan and Schitter 2018). No formal institutional ethics review or exemption was sought; participation followed Prolific’s standard consent, compensation exceeded the platform minimum, and only pseudonymous IDs were collected.

Conclusion

CRUCIBLE demonstrates that annotation rubric calibration can be automated through a coordinator-mediated iterative loop driven by inter-judge agreement, with no human reference labels used during optimization. This paper’s central finding, however, concerns the criterion that decides when the loop has succeeded. Because the weighted agreement statistic the loop optimizes has a floor of 0.667, a threshold of $\tau = 0.80$ is satisfied at 40% unanimity, and the domain that crossed it did so with a chance-corrected agreement of 0.286. A human validation study spanning the full corpus, at double the sample size of our earlier report, does not support the human-alignment dissociation we previously described as preliminary: judge and human reliability rank the three domains identically, and match against human consensus fails to beat majority-class labelling in five of six cells. For trustworthy AI evaluation, the actionable takeaway is that a stopping rule must be defined on a statistic whose scale is known—an automated judge panel reporting agreement above a threshold is making a claim only as strong as the floor of the metric thresholded.

Acknowledgements

The author thanks the Prolific annotators whose judgments constitute the human validation study, and the anonymous ATRACC reviewers, whose questions about the statistical strength of the original comparison prompted the enlarged human study reported here.

References

- Anthropic. 2026. Claude Sonnet 4.6 Model Documentation. Anthropic. Coordinator model for all calibration runs reported here; accessed July 2026.
- Byrt, T.; Bishop, J.; and Carlin, J. B. 1993. Bias, Prevalence and Kappa. *Journal of Clinical Epidemiology*, 46(5): 423–429.
- Chan, C.-M.; Chen, W.; Su, Y.; Yu, J.; Xue, W.; Zhang, S.; Fu, J.; and Liu, Z. 2023. ChatEval: Towards Better LLM-based Evaluators through Multi-Agent Debate. *arXiv preprint arXiv:2308.07201*.
- Cohen, J. 1960. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement*, 20(1): 37–46.
- Das, B. 2026. Signal-to-Noise Ratio Detection in Educational Video Content: A Binary Classification Framework Using Transcript Embeddings. SSRN preprint. Companion technical report; full designs for the diagnostic analyses summarized here.
- Ding, L. 2026. AdaRubric: Task-Adaptive Rubrics for Reliable LLM Agent Evaluation and Reward Learning. *arXiv preprint arXiv:2603.21362*.
- Du, Y.; Li, S.; Torralba, A.; Tenenbaum, J. B.; and Mordatch, I. 2023. Improving Factuality and Reasoning in Language Models through Multiagent Debate. *arXiv preprint arXiv:2305.14325*.
- Fleiss, J. L. 1971. Measuring Nominal Scale Agreement Among Many Raters. *Psychological Bulletin*, 76(5): 378–382.
- Fu, J.; Ng, S.-K.; Jiang, Z.; and Liu, P. 2023. GPTScore: Evaluate as You Desire. *arXiv preprint arXiv:2302.04166*.
- Gwet, K. L. 2008. Computing Inter-Rater Reliability and Its Variance in the Presence of High Agreement. *British Journal of Mathematical and Statistical Psychology*, 61(1): 29–48.
- Hashemi, H.; Eisner, J.; Rosset, C.; Durme, B. V.; and Kedzie, C. 2024. LLM-Rubric: A Multidimensional, Calibrated Approach to Automated Evaluation of Natural Language Texts. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 13806–13834.
- Hinton, G.; Vinyals, O.; and Dean, J. 2015. Distilling the Knowledge in a Neural Network. *arXiv preprint arXiv:1503.02531*.
- Kim, S.; Suk, J.; Longpre, S.; Lin, B. Y.; Shin, J.; Welleck, S.; Neubig, G.; Lee, M.; Lee, K.; and Seo, M. 2024. Prometheus 2: An Open Source Language Model Specialized in Evaluating Other Language Models. *arXiv preprint arXiv:2405.01535*.
- Liu, Y.; Iter, D.; Xu, Y.; Wang, S.; Xu, R.; and Zhu, C. 2023. G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2511–2522.
- OpenAI. 2026. GPT-5.6 Sol Model Documentation. OpenAI. General availability 9 July 2026; used as the non-Anthropic frontier baseline judge, accessed 19 July 2026.
- Palan, S.; and Schitter, C. 2018. Prolific.ac—A Subject Pool for Online Experiments. *Journal of Behavioral and Experimental Finance*, 17: 22–27.
- Rao, D.; and Callison-Burch, C. 2026. Autorubric: A Unifying Framework for Rubric-Based LLM Evaluation on Non-Verifiable Tasks. *arXiv preprint arXiv:2603.00077*.
- Ratner, A.; Bach, S. H.; Ehrenberg, H.; Fries, J.; Wu, S.; and Ré, C. 2017. Snorkel: Rapid Training Data Creation with Weak Supervision. *Proceedings of the VLDB Endowment*, 11(3): 269–282.
- Shen, W. F.; Qiu, X.; Whitehouse, C.; Alazraki, L.; Goel, S.; Barbieri, F.; Willi, T.; Mathur, A.; and Leontiadis, I. 2026. Rethinking Rubric Generation for Improving LLM Judge and Reward Modeling for Open-ended Tasks. *arXiv preprint arXiv:2602.05125*.
- Shi, S.; Wei, R.; Tufano, M.; Cambronero, J.; Cheng, R.; Ivančić, F.; and Rondon, P. 2025. Towards a Human-in-the-Loop Framework for Reliable Patch Evaluation Using an LLM-as-a-Judge. *arXiv preprint arXiv:2511.10865*.
- Siro, C.; Aliannejadi, P.; and Aliannejadi, M. 2026. Learning to Judge: LLMs Designing and Applying Evaluation Rubrics. *arXiv preprint arXiv:2602.08672*.
- Su, E.; Wu, J.; Tang, C.; Wang, L.; Li, P.; Wang, A.; Zhang, J.; Wang, Y.; Meng, Y.; Ma, X.; Tang, S.; and Li, H. 2026. SciIF: Benchmarking Scientific Instruction Following Towards Rigorous Scientific Intelligence. *arXiv preprint arXiv:2601.04770*.
- Thakur, A. S.; Choudhary, K.; Ramayapally, V. S.; Vaidyanathan, S.; and Hupkes, D. 2024. Judging the Judges: Evaluating Alignment and Vulnerabilities in LLMs-as-Judges. *arXiv preprint arXiv:2406.12624*.
- Wei, J.; Wang, X.; Schuurmans, D.; Bosma, M.; Ichter, B.; Xia, F.; Chi, E.; Le, Q.; and Zhou, D. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 24824–24837.
- Zhang, J.; Hsieh, C.-Y.; Yu, Y.; Zhang, C.; and Ratner, A. 2022. A Survey on Programmatic Weak Supervision. *arXiv preprint arXiv:2202.05433*.
- Zheng, L.; Chiang, W.-L.; Sheng, Y.; Zhuang, S.; Wu, Z.; Zhuang, Y.; Lin, Z.; Li, Z.; Li, D.; Xing, E. P.; Zhang, H.; Gonzalez, J. E.; and Stoica, I. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 46595–46623.